

Bayram Bayramov

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Profile

Early-career AI / ML Engineer and Data Scientist with hands-on experience building machine learning and generative AI systems across the full lifecycle—from experimentation to deployment. Built SAR image segmentation models during an internship, developed a production-ready AI food analyzer application, and contributed to data preparation workflows for ASR model evaluation. Experience teaching and mentoring technical concepts, with strong communication skills for bridging AI/ML concepts to diverse audiences. Comfortable working in collaborative, fast-moving environments.

Tech Stack: Python, PyTorch, TensorFlow, Hugging Face Transformers, FastAPI, Docker, PostgreSQL, Git, NumPy, Pandas, Scikit-learn, OpenCV

Experience

DataMinds Azercell, Data Analytics and AI Bootcamp

Baku, Azerbaijan
July 2025 – Aug 2025
2 months

- Built a RAG application and forecasting pipeline during an intensive bootcamp, using vector databases and LLM frameworks
- Gained exposure to MLOps concepts including pipeline orchestration, containerization, and stream processing through additional projects
- Developed foundational understanding of end-to-end ML/AI system design

Kapital Bank, Data Science and AI Intern

Baku, Azerbaijan
Aug 2025 – Aug 2025
1 month

- Contributed to a 4-person labeling team, preparing and validating audio data for an Automatic Speech Recognition (ASR) model
- Assisted in standardizing data preparation workflows to improve consistency and turnaround for model testing
- Collaborated with engineers to align data processing with model requirements, supporting more reliable ASR evaluation

Div Academy, Teacher and Tutor

Baku, Azerbaijan
Oct 2025 – Mar 2026
6 months

- Taught Python, SQL, PyTorch, TensorFlow, FastAPI, and Hugging Face Transformers, translating complex ML/AI and deep learning concepts into accessible practical knowledge
- Provided ongoing technical mentorship and code review, helping learners build skills in model design, training, and deployment
- Developed curriculum materials and coding exercises focused on real-world application

Azercosmos, Machine Learning Intern

Baku, Azerbaijan
Mar 2026 – May 2026
3 months

- Built and trained SAR image segmentation models including U-Net, Attention U-Net, and LRAU-Net in PyTorch, covering the full experimental lifecycle from data preprocessing to evaluation
- Diagnosed model failures through systematic analysis including failure-case visualization, per-region error statistics, and NDVI/NDBI indices
- Developed an evaluation framework that informed data relabeling and model improvements

Projects

AI Food Analyzer

- Built a production-ready web application using FastAPI that identifies ingredients from meal photos and calculates nutritional information via Vision Language Models (OpenAI, Anthropic, Google Gemini) with automatic failover
- Implemented 3-layer caching (VLM, nutrition, HTTP) achieving 500-600x speed improvement and parallel nutrition lookups with asyncio for up to 13x faster processing
- Containerized with Docker and Docker Compose using multi-stage builds reducing image size to ~585MB, with GitHub Actions CI achieving 86% test coverage across 190+ tests
- Integrated PostgreSQL with asyncpg for persistent storage and built both REST API and CLI interfaces with comprehensive OpenAPI documentation
- Designed system architecture with file validation, SHA-256 hashing for caching, TTL-aware cache backends (JSON/SQLite), and structured logging with Loguru

Education

Certificate Program	AI Academy , Artificial Intelligence • Studying machine learning and deep learning with a theory-heavy curriculum, implementing concepts through hands-on coding projects • Built an AI Food Analyzer project applying ML/DL techniques up to CNNs, with the full implementation detailed on GitHub	Baku, Azerbaijan Aug 2025 – Aug 2026
Master	Azerbaijan State University of Economics , Business and Data Analytics • Currently pursuing a Master's degree combining data science with business applications • Coursework includes advanced analytics, data visualization, and quantitative methods	Baku, Azerbaijan Sept 2025 – present
Bachelor	Baku Higher Oil School , Chemical Engineering • Built a strong foundation in mathematics, physics, and computational methods through a rigorous engineering curriculum • Developed analytical problem-solving skills applicable to data science and modeling	Baku, Azerbaijan Sept 2020 – June 2025